

Metabolic Costs of Military Load Carriage over Complex Terrain

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ABSTRACT Introduction: Dismounted military operations often involve prolonged load carriage over complex terrain, which can result in excessive metabolic costs that can directly impair soldiers' performance. Although estimating these demands is a critical interest for mission planning purposes, it is unclear whether existing estimation equations developed from controlled laboratory- and field-based studies accurately account for energy costs of traveling over complex terrain. This study investigated the accuracy of the following equations for military populations when applied to data collected over complex terrain with two different levels of load carriage: American College of Sports Medicine (2002), Givoni and Goldman (1971), Jobe and White (2009), Minetti et al (2002), Pandolf et al (1977), and Santee et al (2003). Materials and Methods: Nine active duty military personnel (age 21 ± 3 yr; height 1.72 ± 0.07 m; body mass 83.4 ± 12.9 kg; VO_2 max 47.8 ± 3.9 mL/kg/min) were monitored during load carriage (with loads equal to 30% and 45% of body mass) over a 10-km mixed terrain course on two separate test days. The course was divided into four 2.5-km laps of 40 segments based on distance, grade, and/or surface factors. Timing gates and radio-frequency identification cards (SportIdent; Scarborough Orienteering, Huntington Beach, CA) were used to record completion times for each course segment. Breath-by-breath measures of energy expenditure were collected using portable oxygen exchange devices (COSMED Sri., Rome, Italy) and compared model estimates. Results: The Santee et al equation performed best, demonstrating the smallest estimation bias (-13 ± 87 W) and lowest root mean square error (99 W). Conclusion: Current predictive equations underestimate the metabolic cost of load carriage by military personnel over complex terrain. Applying the Santee et al correction factor to the Pandolf et al equation may be the most suitable approach for estimating metabolic demands in these circumstances. However, this work also outlines the need for improvements to these methods, new method development and validation, or the use of a multi-model approach to account for mixed terrain.

INTRODUCTION

Military operations often require dismounted personnel to carry heavy loads over long distances.¹ Although loads of 30% body mass or less are recommended to avoid performance decrements,² a previous study of combat operations in Afghanistan showed that the average light infantry rifleman carried over 43 kg of critical equipment during approach marches.³ Although additional equipment increases versatility, heavy load carriage significantly increases energy costs.⁴ Excessive energy expenditure can degrade soldier performance, contribute to thermal strain, as well as lead to physical and cognitive fatigue. Therefore, the ability to accurately predict energy costs is important for effectively planning soldier activities.

Practitioners often rely on predictive equations to estimate the metabolic costs of military load carriage.⁵ Many of the well-established predictive equations were developed from laboratory steady-state exercise protocols that controlled for metabolic rate moderators such as walking speed, treadmill grade, and external load.^{6–9} An inherent assumption is that predictive equations developed from steady-state data can be used to

predict energy costs under dynamic conditions where activities change frequently enough that equilibrium cannot be achieved. Military examples of dynamic conditions include dismounted patrols and approach marches that often require traversing landscapes with shifting gradients and surfaces.¹⁰ As these conditions cannot be realistically replicated in laboratory settings, field research data are necessary to determine whether these predictive equations accurately estimate the metabolic costs of military load carriage over complex terrain. However, there has not yet been a direct comparison between predictive equations using metabolic data collected over complex terrain. These data will provide a basis for realistic expectations regarding the accuracy of metabolic cost predictions for load carriage.

This investigation sought to evaluate how well-established predictive equations estimate the metabolic costs of military load carriage over complex terrain and to identify the most suitable equations for these estimations. Ultimately, the findings of this investigation lay the ground work for future efforts to refine, hybridize, or apply multi-model approaches to improving the accuracy of metabolic cost estimation methods.

MATERIALS AND METHODS

Study Design

Test volunteers completed two marches over a complex terrain course carrying a different external load (either an additional 30% or 45% body mass load). The marches were separated by a minimum of 5 d and the order of the external load conditions

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TABLE I. Predictive Equations from Previous Literature

Reference	Equation
ACSM ¹¹	$(\text{mlO}_2 \cdot \text{kg}^{-1} \cdot \text{min}^{-1}) = 0.1 \cdot S + 1.8 \cdot S \cdot G + 3.5$
Givoni and Goldman ⁷	$(\text{kcal} \cdot \text{hr}^{-1}) = \eta \cdot (M+L) \cdot [2.3 + 0.32 \cdot (S-2.5)^{1.65} + G(0.2 + 0.07 \cdot (S - 2.5))]$
Jobe and White ¹²	$(\text{J} \cdot \text{kg}^{-1} \cdot \text{m}^{-1}) = 20.9 \cdot \tan(G)^2 + 4.18 \cdot \tan(G) + 1.38 \cdot -60 < G < -6$
Minetti et al ²⁰	$(\text{J} \cdot \text{kg}^{-1} \cdot \text{m}^{-1}) = 52.1 \cdot 10^{-3} \cdot \tan(G)^2 + 10.4 \cdot \tan(G) + 2.65 \cdot -6 < G < 60$
Pandolf et al ⁶	$(\text{J} \cdot \text{kg}^{-1} \cdot \text{m}^{-1}) = 280.5 \cdot G^5 - 58.7 \cdot G^4 - 76.8 \cdot G^3 + 51.9 \cdot G^2 + 19.6 \cdot G + 2.5$
Santee et al ⁸	$(W) = 1.5 \cdot M + 2.0 \cdot (M + L) + \eta \cdot (M + L) \cdot (1.5 \cdot S^2 + 0.35 \cdot S \cdot G)$
	$(W) = -\eta \cdot [(G \cdot (M + L) \cdot S) / 3.5 - ((M + L) \cdot (G + 6^2) / M) + (25 \cdot S^2)]$
	*Added to Pandolf et al. ⁶ when $G < 0$

G, grade (% for Givoni and Goldman, Pandolf, and Santee, decimal for rest); L, external load (kg); M, body mass (kg); S, speed (m/min for ACSM, km/h for Givoni and Goldman, m/s for rest).

was randomized. Six predictive equations^{6-8,11-13} (Table 1) were used to estimate metabolic rate over course segments and evaluated against measurements from a portable oxygen uptake monitor system (COSMED K4b² (K4) COSMED Sri., Rome, Italy). The Santee et al equation used in this investigation includes a multiplication sign following the S^2 term instead of the minus sign that was erroneously included in the original published equation.

Volunteers

Nine (eight males and one female) active duty military personnel (mean $\pm$ SD; age, 21 ± 3 yr; height, 1.72 ± 0.07 m; body mass, 83.4 ± 12.9 kg; VO_2 max, 47.8 ± 3.9 ml/kg/min) volunteered to participate in this investigation. All volunteers were medically cleared before participation. Selection criteria included general good health as determined during clearance procedures, passing their most recent Army Physical Fitness Test, and no recent history of musculoskeletal injury or other pre-existing conditions that would preclude their participation in the study. The study was approved by both the Scientific Review Committee and Institutional Review Board at the U.S. Army Research Institute of Environmental Medicine (USARIEM) (Natick, MA). The USARIEM Institutional Review Board was responsible for regulatory oversight of the investigation.

Procedures

Complex Terrain Course

All testing was conducted on the Natick Soldier System Center Fitness Trail in Natick, MA (Fig. 1). The complex terrain course (~2.5 km) was divided into 40 segments based on distance (60.1 ± 45.6 m), grade ($0.4 \pm 6.6\%$), and surface factors. Terrain factors of 1.0 and 1.2 were assigned for paved road (49.9% of total distance) and dirt road (50.1% of total distance) segments of the course in accordance with the recommendations of Richmond et al¹⁰. The altitude of the course ranged between 42 and 52 m above sea level. Two Remote Automated Weather Stations collected air temperature and other weather parameters during data collection (WeatherHawk 510; WeatherHawk, and Campbell CR3000;

Campbell Scientific Instruments, Logan, UT). In addition, natural wet-bulb, dry-bulb, and black globe temperatures, used to calculate the wet bulb-globe temperature index, were collected with the CR3000 and a QuesTemp34 wet bulb-globe temperature monitor (Quest Technologies, Oconomowoc, WI). Average environmental data included the following: wet-bulb temperature ($16.9 \pm 4.4^\circ\text{C}$), dry-bulb temperature ($21.1 \pm 4.9^\circ\text{C}$), globe temperature ($29.5 \pm 7.5^\circ\text{C}$), wet-bulb globe temperature ($19.8 \pm 4.8^\circ\text{C}$), and relative humidity ($52 \pm 11\%$).

Familiarization

During initial pack and load fitting, volunteers were familiarized with carrying both the 30% and 45% loads over a short distance (50–150 m) on a level surface. Volunteers participated in a familiarization session for exposure to testing equipment, procedures, and the complex terrain course before data collection. Upon arrival, volunteers were fitted to a portable oxygen uptake (VO_2) monitor system (COSMED K4b² (K4) COSMED Sri., Rome, Italy). The body-mounted elements of the system consisted of a Portable Unit, a battery, a face mask covering the nose and mouth, mounting harness, and connecting tubes and cables. Each volunteer then walked one clockwise lap around the complex terrain course while accompanied by a research team member. Volunteers were not fitted with other instrumentation during this familiarization.

Test Visits

Each test visit began with preliminary measurements and instrumentation before a ~150 m warm-up walk without an external load. Subsequently, volunteers were fitted with the portable VO_2 monitor and donned either the 30% or 45% load. The 30% load included the Army Combat Uniform uniform with the Army Combat Shirt, personal boots, with a non-functional replica rifle, body armor, water, and additional load to simulate munitions, but no rucksack. Additional weight was added to the vest pouches if the Clothing and Individual Equipment described above weighed less than 30% body mass. To be relevant to military training and operations, all volunteers carried at least the equivalent of a fighting or combat load, which includes body armor. The 45% load also included a Modular Lightweight Load Carrying Equipment

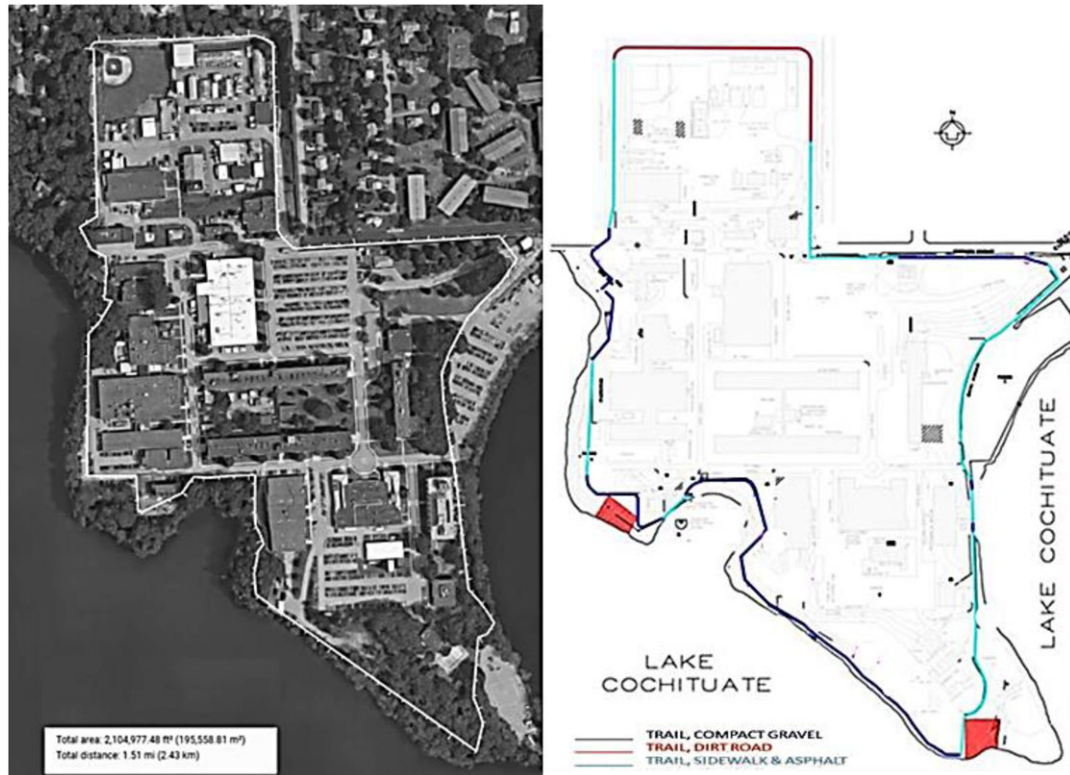


FIGURE 1. Overview of complex terrain course (Natick Soldier System Center Fitness Trail, Natick, MA).

frame rucksack; additional weight was placed into the rucksack to increase the total load to 45%. After standing for 10 min to collect baseline VO_2 data, volunteers were then required to complete a minimum of two and a maximum of four laps around the complex terrain course as fast as possible without running, jogging, or shuffling. Volunteers were provided 5-min rest periods between laps during which they were allowed to sit down, temporarily remove their masks, and drink water. Each volunteer was provided a radio-frequency identification tag (SportIdent; Scarborough Orienteering, Huntington Beach, CA), which communicated with timing gates (SportIdent; Scarborough Orienteering, Huntington Beach, CA) placed between segments to record completion times and speeds along the course. Volunteers completed laps 1 and 3 in the clockwise direction and laps 2 and 4 in the counterclockwise direction. Volunteers completed an average of 3.6 ± 0.8 laps per test session with four trials terminated early due to threshold constraints on environmental exposures and in one case due to a significant increase in core temperature and other indications of heat strain. However, all volunteers met the minimum requirement and completed at least one lap in each direction during a given test session.

Statistical Analyses

All data are displayed as mean $\pm$ standard deviation (SD) and analyzed using RStudio (Version 0.98.1056; RStudio, Inc, Boston, MA). Metabolic rate estimations and observations

were compared over the overall course as well as individual course segments. Predictive equation performance was evaluated by the bias, residual SD, root mean square error (RMSE), and Pearson's correlation coefficient (r). The external load was added to body mass for the physical activity component of the equations that did not specifically account for external loading (ACSM, Jobe et al, Minetti et al) in order to account for the linear increase in metabolic rate with additional load.¹⁴ Resting metabolic rates ($\text{mL VO}_2/\text{kg}/\text{min}$) were calculated for the Jobe et al and Minetti et al equations using the following equations from Schofield et al¹⁵:

$$\text{Resting VO}_2 (\text{Male, 18–30 yr}) = 2.177 + (100.055 \cdot M^{-1})$$

$$\text{Resting VO}_2 (\text{Female, 18–30 yr}) = 2.141 + (70.343 \cdot M^{-1})$$

RESULTS

Figure 2 displays mean VO_2 per segment over the first two laps for the 30% and 45% body mass load tests. The mean VO_2 over the entire course was $22.0 \pm 4.4 \text{ mL O}_2/\text{kg}/\text{min}$. Table 2 summarizes the performance of each predictive equation in estimating metabolic rates for load carriage over complex terrain. Nearly all predictive equations produced estimates that were lower compared with the observed mean metabolic rate of all segments ($655 \pm 176 \text{ W}$) as well as the observed mean across all courses completed ($638 \pm 143 \text{ W}$). The Pandolf et al equation had comparable accuracy when

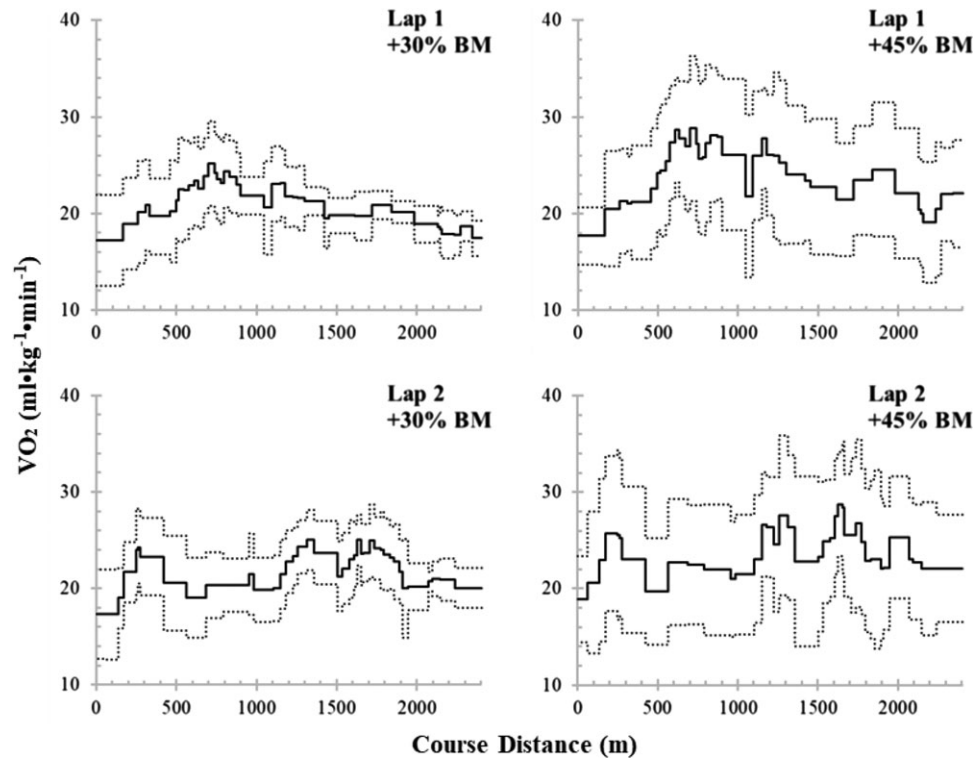


FIGURE 2. Mean oxygen consumption (VO_2) per segment over the first two laps for the complex terrain marches with 30% and 45% body mass loads. Solid lines, mean VO_2 per segment; dotted lines, mean $\pm$ SD.

TABLE II. Predictive Equation Performance (W)

Grade	Equation	Segment			Course		
		Bias $\pm$ SD	RMSE	r	Bias $\pm$ SD	RMSE	r
All	ACSM ¹¹	-163 ± 507	532	0.22	-140 ± 104	174	0.70
	Givoni and Goldman ⁷	-127 ± 359	381	0.32	-111 ± 90	143	0.78
	Jobe et al ¹²	-87 ± 181	201	0.46	-73 ± 100	124	0.73
	Minetti et al ²⁰	-69 ± 262	271	0.39	-74 ± 100	124	0.73
	Pandolf et al ⁶	-72 ± 463	468	0.29	-51 ± 88	102	0.79
	Santee et al ⁸	-38 ± 313	315	0.44	-13 ± 87	99	0.80
>0%	ACSM ¹¹	112 ± 395	410	0.39	53 ± 107	120	0.69
	Givoni and Goldman ⁷	57 ± 295	300	0.47	15 ± 99	105	0.77
	Jobe et al ¹²	-27 ± 162	164	0.57	-19 ± 101	102	0.71
	Minetti et al ²⁰	49 ± 283	288	0.44	20 ± 101	103	0.71
	Pandolf et al ⁶	-92 ± 138	166	0.65	102 ± 102	144	0.78
	Santee et al ⁸	-92 ± 138	166	0.65	175 ± 370	409	0.46
<0%	ACSM ¹¹	-435 ± 456	630	-0.03	-333 ± 115	352	0.68
	Givoni and Goldman ⁷	309 ± 322	447	0.11	-221 ± 85	237	0.80
	Jobe et al ¹²	-146 ± 181	232	0.32	-127 ± 101	162	0.73
	Minetti et al ²⁰	-186 ± 173	254	0.34	-168 ± 103	197	0.74
	Pandolf et al ⁶	-314 ± 412	520	0.06	-202 ± 87	219	0.80
	Santee et al ⁸	-98 ± 147	177	0.56	-75 ± 83	92	0.81

r , Pearson's correlation coefficient.

predicting metabolic rates per course with and without the corrective factor of Santee et al (Bias, -51 ± 88 W and -13 ± 87 W, respectively). However, the Santee et al equation had a lower RMSE than the Pandolf et al equation when estimating overall metabolic rates per course (99 W vs. 102 W).

DISCUSSION

The most salient finding of this investigation was that each of the predictive equations underestimated the metabolic costs of military load carriage over complex terrain. The adjusted Pandolf et al equation had the most accurate metabolic rate

estimates among the six equations evaluated. However, these results clearly demonstrate the need for improvements to these methods, replacement with new methods, or the use of a multi-model approach to account for mixed terrain.

Metabolic rate estimates from the predictive equations were considerably lower than the metabolic rates measured by the COSMED K4b². Although many studies have shown Pandolf et al to be an accurate predictor of metabolic costs in the lab,¹⁶ some have highlighted specific discrepancies linked to external load configurations (e.g., heavy ballistic suits).¹⁷ This brings into question some of the other significant contributors to individualized effects, such as different proportional loading (% body mass), fitness levels (% VO₂ max), hydration status, or sex. It is unlikely that metabolic rates were higher than estimated values due to unfamiliar or unfit test volunteers as they were experienced with heavy load carriage and were aerobically fit (VO₂ max, 47.8 ± 3.9 mL/kg/min). It is also possible that certain course segments may have been too brief to approximate the steady-state metabolic rates predicted by the equations. Alternatively, just as heart rates remain elevated after cresting a steep uphill grades, it is highly possible that the more physically demanding segments caused persistent metabolic rate elevations over subsequent segments. Another complication to this is the fact that the length of individual breathes vary from ~3–10 sec and it can take less than or only slightly more than this time for an individual to traverse the shorter trail segments, it was as expected that some lag or overshooting would occur in metabolic responses.

The metabolic rates measured by the COSMED K4b² may have been higher than true values. McLaughlin et al¹⁸ found that the COSMED K4b² overestimated VO₂ consumption during cycle ergometer exercise by 50–200 W when compared with the Douglas bag method. Similarly, Duffield et al¹⁹ found that the COSMED K4b² system overestimated VO₂ during intermittent treadmill exercise compared with a laboratory metabolic cart but noted that the COSMED K4b² system demonstrated satisfactory test–retest reliability. In contrast, Schrack et al²⁰ found that VO₂ measurements by the COSMED K4b² were not significantly different ($p = 0.25$) but highly correlated ($r = 0.96$) to a traditional, stationary gas exchange during steady-state walking.

A consistent offset by the COSMED K4b² would call into question whether the Pandolf et al and Santee et al equations, which had the smallest bias and RMSE, were the best predictors of the true metabolic rate. The correlation between estimates and observations would be a better indicator of predictive equation performance in this case. However, the Pandolf et al and Santee et al equations also had the highest correlations with the observed metabolic rate for the course ($r = 0.79$ and 0.80 , respectively) as well as uphill ($r = 0.78$ for each) and downhill segments ($r = 0.80$ and 0.81 , respectively). Consequently, the findings of this investigation support the use of the Pandolf et al equation with the correction factor from Santee et al as most suitable for estimating

metabolic demands when traversing complex terrain courses.

Combinations of rational and empirical models have been used to produce more diverse and accurate predictions for physiological prediction modeling.²¹ This approach could conceivably be outlined from the current work by simple model selection based on grade alone, where methods are applied based on the best performing equation for level or uphill grades ($\geq 0\%$) and for downhill ($<0\%$). For example, applying the equation from Jobe et al for positive-grade segments (RMSE = 102 W) and Santee et al for negative-grade segments (RMSE = 92 W) should provide more accurate estimation for the overall energy expenditure for a known course. As the terrain features themselves have significant impact on the energy demands, grade alone may be a simplistic concept but could be extended to include several models in this multi-model approach specifically targeted to more defined terrain and surface features.

The present investigation has a few notable limitations. For instance, the sample size was only nine volunteers with a single female volunteer. However, it should be noted that comparable samples were used in many of the original load carriage studies²² including the datasets used to validate the Minetti et al¹³ ($n = 10$), Pandolf et al⁶ ($n = 6$), and Santee et al⁸ ($n = 16$) equations. Although all volunteers were directed to attempt four laps, not all were able to complete all of the laps during each test session. This was due to threshold constraints on environmental exposures ($n = 4$) and significantly increased core temperature ($n = 1$). However, all volunteers completed at least one lap in each direction during a given test session.

CONCLUSION

Current predictive equations underestimate the metabolic costs of military load carriage over complex terrain. From this study, a combination of the Pandolf et al equation with the correction factor from Santee et al is the most suitable method for estimating metabolic demands over complex terrain. However, the disagreement between predicted and observed values reported in this article indicates a need for improvement using possible methods such as a multi-model approach.

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